

MAT 307 HW01

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Due 9/12

Problem 1. Using the dot product, prove the Pythagorean theorem.

Solution. Let $\triangle ABC$ be a triangle with side lengths $BC = a$, $CA = b$, $AB = c$. We see that

$$AB^2 = \vec{AB} \cdot \vec{AB} = (\vec{AC} + \vec{CB}) \cdot (\vec{AC} + \vec{CB}) = AC^2 + BC^2 + 2\vec{AC} \cdot \vec{CB}.$$

That is,

$$c^2 = a^2 + b^2 + 2\vec{AC} \cdot \vec{CB}.$$

This immediately shows us that $a^2 + b^2 = c^2$ if and only if $2\vec{AC} \cdot \vec{CB} = 0$, i.e. AC and BC are perpendicular. $\square$

Problem 2. Let $\vec{u}$ and $\vec{v}$ be vectors in $\mathbb{R}^2$ and recall that the area of the triangle defined by $\vec{u}$ and $\vec{v}$ is given by $A(\vec{u}, \vec{v}) = \frac{1}{2}|\vec{v}^\perp \cdot \vec{u}|$, where $\vec{v}^\perp$ is the counterclockwise rotation of $\vec{v}$ by 90° .

- (a) Show that $(\vec{v}^\perp \cdot \vec{u})^2 = \|\vec{u}\|^2 \|\vec{v}\|^2 - (\vec{u} \cdot \vec{v})^2$. Explain what this means geometrically.
- (b) Use this fact to establish Heron's formula.
- (c) Show that, for fixed side lengths a and b , the triangle with the largest area is right.
- (d) Show that, for a fixed perimeter, the triangle with the largest area is equilateral.

Solution. The proof of (a) is obvious. One way to think about this geometrically is to see that the projections of $\vec{u}$ on $\vec{v}$ and $\vec{v}^\perp$ are the legs of a right triangle with hypotenuse $\vec{u}$; this can be seen by the equivalent equation

$$\left\| \frac{\vec{v} \cdot \vec{u}}{\|\vec{v}\|^2} \vec{v} \right\|^2 + \left\| \frac{\vec{v}^\perp \cdot \vec{u}}{\|\vec{v}^\perp\|^2} \vec{v}^\perp \right\|^2 = \|\text{Proj}_{\vec{v}} \vec{u}\|^2 + \|\text{Proj}_{\vec{v}^\perp} \vec{u}\|^2 = \|\vec{u}\|^2$$

due to the Pythagorean theorem.

(b) is an algebra bash.

(c) follows from (a), since we want to minimize the square of dot product of $\vec{u}$ and $\vec{v}$, which happens when the dot product is zero and thus the vectors are orthogonal.

(d) Jensen carry ngl $\square$

Problem 3. (a) Let $\vec{u}, \vec{v} \in \mathbb{R}^2$ be non-parallel vectors. Describe the set of vectors $s\vec{u} + t\vec{v}$ where $s + t = 1$. What about if s and t are also nonnegative?

- (b) Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ be non-coplanar vectors. Describe the set of vectors $r\vec{u} + s\vec{v} + t\vec{w}$ where $r + s + t = 1$. What happens if r, s, t are nonnegative?

Solution. (a): Eliminate t by parameterizing as $\vec{v} + s(\vec{u} - \vec{v})$ to see that this set is the line through the endpoints of the vectors; if $0 \leq s \leq 1$ then it's just the segment between the endpoints of the two vectors.

(b): Same idea, but with "plane" instead of "line" and "triangle" instead of "segment." $\square$

Problem 4. Four vectors are erected perpendicularly to the four faces of an arbitrary tetrahedron. Each vector is pointing outward and has length equal to the area of the corresponding face. Show that these vectors sum to the zero vector.

Solution. Scale the vectors by a factor of two; this doesn't affect whether or not they sum to zero, so it's equivalent to show that the scaled vectors sum to zero.

Anchor one of the vertices of the tetrahedron to the origin and let $\vec{a}$, $\vec{b}$, and $\vec{c}$ be the vectors along the edges of the vertices emanating from the origin, labeled so that $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$, and $\vec{c} \times \vec{a}$ are all pointing outward from the tetrahedron. Then, the four vectors are the aforementioned cross products along with the product $(\vec{c} - \vec{a}) \times (\vec{b} - \vec{a})$. The desired sum is then

$$\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} + (\vec{c} \times \vec{b} - \vec{c} \times \vec{a} - \vec{a} \times \vec{b} + \vec{a} \times \vec{a}),$$

using the fact that the cross product distributes over vector addition. Then, since the cross product is anticommutative and the cross product of a vector with itself is zero (area of parallelogram spanned by $\vec{a}$ and $\vec{a}$ is zero), we simplify as

$$\vec{b} \times \vec{c} + \vec{c} \times \vec{b} + \vec{a} \times \vec{a} = \vec{b} \times \vec{c} - \vec{b} \times \vec{c} + \vec{0} = \vec{0}.$$

□

Problem 5. (a) Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be three vectors that are not coplanar, i.e. $\alpha\vec{u} + \beta\vec{v} + \gamma\vec{w} = \vec{0}$ implies that $\alpha = \beta = \gamma = 0$. Show that $\vec{u} \times \vec{v}$, $\vec{v} \times \vec{w}$, and $\vec{w} \times \vec{u}$ are not coplanar.

(b) Suppose a , b , and c are reals. Express the intersection point of planes

$$\vec{u} \cdot (x, y, z) = a, \quad \vec{v} \cdot (x, y, z) = b, \quad \vec{w} \cdot (x, y, z) = c$$

as a linear combination of $\vec{u} \times \vec{v}$, $\vec{v} \times \vec{w}$, and $\vec{w} \times \vec{u}$.

Solution. (a) is easy to do by contradiction: if the cross products are coplanar, then one of them (say $\vec{v} \times \vec{w}$) can be written as a linear combination of the other two. The other two are both orthogonal to $\vec{u}$, so $\vec{v} \times \vec{w}$, being a linear combination of the other two, is also orthogonal to $\vec{u}$; however, this means that $\vec{u}$, $\vec{v}$, and $\vec{w}$ are coplanar, contradiction!

(b) Plugging in

$$(x, y, z) = \alpha\vec{v} \times \vec{w} + \beta\vec{w} \times \vec{u} + \gamma\vec{u} \times \vec{v},$$

the three plane equations become

$$\begin{aligned} \vec{u} \cdot (\alpha\vec{v} \times \vec{w} + \beta\vec{w} \times \vec{u} + \gamma\vec{u} \times \vec{v}) &= \vec{u} \cdot (\alpha\vec{v} \times \vec{w}) = \alpha(\vec{u} \cdot (\vec{v} \times \vec{w})) = a \\ \vec{v} \cdot (\alpha\vec{v} \times \vec{w} + \beta\vec{w} \times \vec{u} + \gamma\vec{u} \times \vec{v}) &= \vec{v} \cdot (\beta\vec{w} \times \vec{u}) = \beta(\vec{v} \cdot (\vec{w} \times \vec{u})) = b \\ \vec{w} \cdot (\alpha\vec{v} \times \vec{w} + \beta\vec{w} \times \vec{u} + \gamma\vec{u} \times \vec{v}) &= \vec{w} \cdot (\gamma\vec{u} \times \vec{v}) = \gamma(\vec{w} \cdot (\vec{u} \times \vec{v})) = c \end{aligned}$$

after distributing the dot product and noting that $\vec{x}$ is perpendicular to the cross product of $\vec{x}$ with any other vector. Then,

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \vec{v} \cdot (\vec{w} \times \vec{u}) = \vec{w} \cdot (\vec{u} \times \vec{v}) = D$$

is the signed area of the parallelepiped formed by $\vec{u}$, $\vec{v}$, $\vec{w}$, and we obtain $(\alpha, \beta, \gamma) = (a/D, b/D, c/D)$. □

Problem 6. Consider a pyramid in $\mathbb{R}^n$ with vertices at the origin O and $\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n$ where the $\hat{e}_i$ form the standard Euclidean basis.

- (a) Find the coordinates of the point C in the base B that is equidistant from each of the $\hat{e}_i$ and compute $\|\vec{OC}\|$.
- (b) Use Cauchy to show that C is the closest point of B to O .
- (c) Compute the angle θ between $\vec{OC}$ and any $\vec{OV}_i$ where V_i is the vertex corresponding to $\hat{e}_i$. What happens to θ as n goes to infinity?

Solution. (a): Since C is the intersection of the perpendicular bisectors of the edges of B (within the plane of B), it is unique if it exists. I claim that $C = (1/n, 1/n, \dots, 1/n)$ is the average of $\hat{e}_1, \hat{e}_2, \dots, \hat{e}_n$; this is trivial to check by just computing the length of each CV_i .

(b): Let $\vec{v}$ be a vector with endpoint in B and $\vec{c} = \vec{OC}$. Note that $\|\vec{c}\| = \sqrt{(1/n)^2 \cdot n} = \frac{\sqrt{n}}{n}$. Cauchy-Schwarz shows that

$$\frac{\|\vec{v}\|}{\sqrt{n}} = \|\vec{v}\| \|\vec{c}\| \geq |\vec{v} \cdot \vec{c}| = \left| \frac{x_1}{n} + \dots + \frac{x_n}{n} \right| = \frac{1}{n},$$

which means that $\|\vec{v}\| \geq \frac{\sqrt{n}}{n} = \|\vec{c}\|$ as desired.

(c): We know that

$$\frac{\vec{OC} \cdot \vec{OV}_i}{\|\vec{OC}\| \|\vec{OV}_i\|} = \cos \theta$$

where θ is the angle we are trying to find. The left-hand side evaluates to

$$\frac{(1/n, 1/n, \dots, 1/n) \cdot (0, \dots, 0, 1, 0, \dots, 0)}{\sqrt{(1/n)^2 \cdot n} \cdot 1} = \frac{1/n}{\sqrt{n}/n} = \frac{\sqrt{n}}{n},$$

so $\theta = \arccos(\sqrt{n}/n)$. Since $\sqrt{n}/n$ goes to zero from the positive direction as n grows large, θ approaches $\arccos(0) = \pi/2$. $\square$